HRISHIKESH M. KAKADE

Firmware & Embedded Software Engineer | UAV / Drone Systems | IoT

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• Amravati, Maharashtra

PROFESSIONAL SUMMARY

Electronics & Telecommunication Engineering student with hands-on industry experience in UAV swarm firmware, RTK GPS setup, Li-ion battery development, PCB design, and IoT embedded systems. Published patent holder in UAV-based flood mapping. Proficient in C/C++, Python, STM32, ESP32, ROS2 Humble, and Pixhawk-based flight systems. Passionate about building reliable embedded and drone solutions for real-world impact.

EXPERIENCE

Embedded Firmware Intern — *Airbotix Technology, Gurugram, Haryana* Jul 2026 – Present

- Gained hands-on experience developing and debugging embedded firmware in a Linux-based development environment, including toolchain setup, build systems, and serial/log-based debugging.
- Worked hands-on with DroneBridge for UAV telemetry and communication link configuration, gaining practical exposure to wireless data link setup between ground control and flight systems.
- Built hands-on experience with the ESP32-C6 module and the ESP-IDF framework, including firmware development, peripheral configuration, and flashing/debugging workflows.

Intern — Hardware, Firmware & Process Automation — *ideaForge Technology Ltd., Navi Mumbai* Nov 2025 – Mar 2026

- Swarm Drone Firmware: Performed end-to-end firmware installation, configuration, and validation on swarm UAV units for coordinated multi-drone operations.
- RTK GPS Setup: Configured RTK (Real-Time Kinematic) GPS systems achieving centimeter-level positioning accuracy across swarm drone fleets.
- Li-ion Battery Development: Contributed to UAV battery pack development — cell selection, assembly validation, charge-discharge cycle testing, and safety compliance.
- ESD Mitigation: Analyzed and resolved ESD failures in UAV subsystems; developed hardware protection documentation for production teams.
- Automation: Built Python scripts to automate BOM auditing, cutting hardware review cycle time by 30%.

Software Development Intern — IoT — *Adolf Solution OPC Pvt. Ltd., Pune* Jun 2025 – Jul 2025

- Developed IoT TCP/IP communication system for cross-network device messaging using ESP32; built remote light control system with real-time Wi-Fi feedback.

Research Associate — *Vihang Drone Club* 2023 – 2026

- Completed 5+ drone and embedded projects: RC planes, obstacle-avoidance cars, UAV mapping systems. Tools: Linux, Pixhawk, Mission Planner, KiCad.

PROJECTS

UAV-Based Flood Mapping & Early Warning System — *Python, ArcGIS, ML, Pixhawk, Mission Planner*

- Published patent (App No. 202523064402): UAV pipeline with ArcGIS flood simulation, ML-based flood prediction, and web portal — fully completed and demonstrated.
- Generated 3D terrain maps from UAV imagery; deployed flood risk-zone overlays with ML prediction models.

Drone PCB — Integrated ESC & Power Distribution — *KiCad, Schematic & Layout*

- Designed and completed a multi-layer drone PCB with integrated ESC, regulated 5V/12V BEC outputs, optimized EMI shielding, and high-current routing.

Line Follower Robot — *C++, IR Sensors, PID Control, Motor Driver*

- Designed, built, and tested an autonomous line-following robot with PID control logic, real-time IR sensor feedback, and differential motor steering — fully operational.

IoT Smart Light Control System — *ESP32, Arduino C++, HTTP, Wi-Fi, HTML/CSS/JS*

- Completed a browser-based remote light control system with real-time ON/OFF state feedback and acknowledgment confirmation via ESP32 Wi-Fi server.

ROS 2 Closed-Loop Robot Controller — *Python, ROS 2, rqt_graph*

- Closed-Loop Control: Developed a ROS 2 Python node utilizing dynamic pose feedback to calculate real-time closed-loop motion states.
- Autonomous Avoidance: Programmed boundary-checking logic to publish velocity commands for obstacle avoidance, verified via rqt_graph.

TECHNICAL SKILLS

Embedded/HW: C, C++, Python | ESP32, ESP32-C6, ESP-IDF, Arduino, STM32, Pixhawk | KiCad (Schematic & PCB Layout)
Robotics/UAV: ROS2 Humble, Mission Planner, Pixhawk FC, ArcGIS, UAV Imagery, Swarm Systems, RTK GPS, DroneBridge
IoT & Comms: TCP/IP, HTTP, Wi-Fi, UART, GPIO | Sensor integration, Remote control systems
Automation: Python scripting, BOM auditing, UiPath RPA (Certified), Process Automation
Tools: VS Code, Linux, KiCad, Google Colab, AI Tools, Mission Planner

EDUCATION

BE Electronics & Telecommunication Engineering — CGPA 6.54/10 Nov 2022 – June 2026
Shri Sant Gajanan Maharaj College of Engineering, Shegaon, Maharashtra

ACHIEVEMENTS & CERTIFICATIONS

- Patent Published: UAV Flood Mapping & Early Warning System (App No. 202523064402, 2025)
- Reasearch Paper: The Future of Drones in Disaster Management: A New Era of Emergency Response (May 2025)
- Top 15 — State-Level Aavishkar 2k25 (Color Coat). Top 3 in 5+ Project ExPO competitions.
- Certifications: UiPath RPA Developer (Udemy) | PCB Design using Autodesk Eagle (Udemy)
- T&P Co-ordinator & Research Associate, Vihang Drone Club (2023–26)